

Pain: the clinical aspects

with EBONIE RIO

*Understanding pain biology changes the way people think about pain,
reduces its threat value and improves their management of it.*

David Butler & Lorimer Moseley, *Explain Pain*

Chapter 5 provided the complex foundational elements of pain science. In this chapter, we share how clinicians can apply this knowledge to better manage patients.¹ Being a clinician would be straightforward if the location of pain always revealed the precise area of pathology and pain had a tissue basis. But this is commonly not the case!

It is useful to consider the contributions to a patient's pain. For example, is there peripheral sensitisation? Is there central sensitisation? Is there referral? Are there cortical contributions? Are there changes in cortical representations? No matter what the answer to these questions, pain is a sign that the brain is urging protection. Identifying the relative importance of different contributions allows you to target your treatment accordingly.

Let's consider a patient with longstanding back pain. Historically, clinicians developed a list of differential diagnoses to classify injury or disease. In this case the options include muscle strain, disc injury or facet joint injury. Working from such a list of differential diagnoses relies on several assumptions. First, that knowledge of the affected structure is important to the assessment and diagnosis. Second, that it is possible to accurately confirm the diagnosis (i.e. that a gold standard exists). Third, that making a tissue diagnosis will inform treatment. Contrary to these 1980s opinions, none of those assumptions appear correct today.

A complementary approach is to consider an overarching classification (e.g. 'the patient had an episode of back pain') that focuses on *the pain experience and contributing factors* as a distinct clinical entity. With this

foundation, the clinician can add information about the state of specific anatomical structures and determine the best treatment approach (which may change over time).

This broader perspective of painful events, which encourages assessment by body region and type of threat, is based on the neuroscience that body movements are represented on motor homunculi and body surfaces are represented on sensory homunculi (Fig. 6.1). While the chapters of this book are arranged to permit region-specific assessment and treatment of what we consider predominantly nociceptive-driven (or input-dominated) presentations, this chapter aims to help clinicians to know when to consider centrally dominated pain in their assessment and management. So, what is meant by 'input-dominated pain' and 'centrally dominated pain'?

INPUT-DOMINATED PAIN

Let's begin with the most common type of pain that brings the patient to a clinician in this field—pain from a musculoskeletal injury. This is 'input-dominated pain' and it is distinguished from 'centrally dominated pain' (discussed in the next section). When learning about concepts like this, it is helpful to artificially distinguish ends of a spectrum. Of course input-dominated and centrally dominated pains occur together, but let's put them in very distinct boxes to start with. In this context, 'input dominated' is a term used to get the clinician to appreciate the nociceptive contribution. The input is the sensory information from the periphery that has stimulated a nociceptor (a receptor capable of transducing and encoding noxious stimuli).

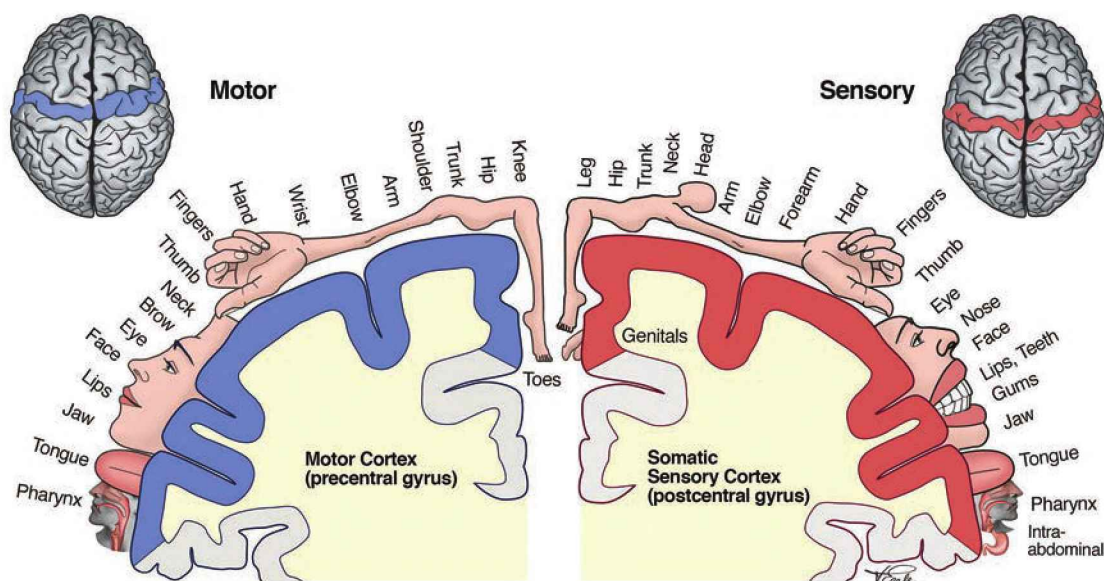


Figure 6.1 Motor and sensory homunculi

Pain and musculoskeletal tissues

Pain is what makes a person go to see a clinician. Acute incidents or injuries are generally associated with nociceptive sensory inputs from tissues. Many patients will report that their pain has affected their function in sporting, occupational or everyday activities. The clinician should evaluate the status of musculoskeletal structures as per a normal clinical encounter (see Chapter 14) and consider the following:

1. joints, including ligaments
2. bones
3. muscles and fascia
4. tendons
5. neural structures (see below).

In acute states, pain associated with nociceptive contributions is thought by clinicians to be fairly closely linked with injury and the repair process. As explained in Chapter 5, the relationship between tissue status (damage), afferent impulses (i.e. impulses perceived as noxious), and pain is not predictable or linear. Afferent impulses may overemphasise the danger associated with an injury at first (e.g. severe pain of an ankle sprain), but this can quickly diminish. The patient can be pain-free at 2 weeks post-injury, but not fully healed for 6 weeks. It is important to avoid thinking of pain presentations purely in terms of the length of time of symptoms. An example of this is pain associated with the patellar tendon (see Chapter 36), where pain is intimately linked with patellar tendon load, implying a nociceptive contribution

even though pain can persist for years. In this example, the pain experience is of course modulated (as all pain is an output), yet there appears evidence of a nociceptive contribution.

Table 6.1 demonstrates examples of nociceptive pain descriptors and clinical examples.

PRACTICE PEARL

It is essential that the clinician eliminates sinister causes such as cancer (see Chapter 7).

Less well characterised input-dominated pain—referred pain

Referred pain is an example of nociceptive input-dominated pain. It can be thought of as an error in the brain's processes that determine where, exactly, it should hurt. That is, the location of the dangerous event is 'misread' by the brain. An example of referred pain is pain in the neck or left arm when it is the heart that is the structure under threat. Another common example is leg pain when nociceptive input arises from a structure in the back or the ascending projections normally associated with that structure. An extreme case of referred pain is 'phantom pain' where pain appears to be localised in a limb that is no longer part of the body (e.g. after amputation). This is also a good example of the discrepancy between apparent input and the experience of pain.

Table 6.1 Clinical examples of terminology and tissue-based examples of pain with nociceptive contributions²

Nociceptive pain descriptors	Tissue-based examples
Patients describe an incident that fits with anatomical location	Ankle sprain may injure ligaments, fascia, muscles, tendon, bone and neural structures via mechanical, ischaemic and/or inflammatory processes
Patient reports clear aggravating and easing factors as well as consistent and clear pain reproduction on movement (or testing) of target tissues	Testing of the involved ligaments reproduces pain
Pain usually eases as the injury settles	Initially someone may present with a limp and this reduces over time, perhaps with appropriate treatment that may include activity modification, ice, compression and optimal loading (graduated exercise progression)

Four case scenarios where referred pain is a common culprit!

- A patient presents with a long history of intermittent, dull, occipital headaches. The patient is thoroughly investigated for eye problems and the presence of intracranial pathology. All tests are normal.
- A patient presents with a history of an ache in the right shoulder that is difficult to localise and is associated with pain on the medial aspect of the upper arm. There is some neck stiffness and tightness in the trapezius muscle.
- A 35-year-old executive complains of episodes of sharp, left-sided chest pain related to activity. The patient has already undergone extensive cardiac investigations that were all normal.

- A young athlete presents with a history of recurrent episodes of buttock and hamstring pain. There is no history of an acute tear and the patient describes the pain as deep-seated and dull with occasional sharp cramping in the hamstring. Examination of the hamstring shows some mild tenderness but full stretch and strength.

These clinical presentations are common in sport and exercise medicine practice. All of the patients are likely to be experiencing referred pain and treatment needs to take this into account.

It is important to keep in mind that there is always a reason for pain (response to perceived threat). Because pain is the end point of a complex danger-relevant processing system, it means that when the tissue that hurts seems normal, we may need to look elsewhere rather than presume the patient is not in pain.

Consider that the brain is wrongly locating a dangerous event—that there is a noxious contributor in an area that does not hurt (and the pain is felt at a remote distance to the noxious contributor). There are common patterns and mechanisms of referred pain. Two of these are ‘radicular referred pain’ and ‘somatic referred pain’. Being familiar with these will stand the young clinician in very good stead. You will make many diagnoses and help many patients who your colleagues will consider ‘difficult’ or even ‘crazy’. (We don’t condone terms like ‘crazy’ about patients, but patients have reported to us that they were told this or they felt crazy, so the term is used here to emphasise the burden that undiagnosed referred pain can have.)

Radicular pain

The concept of radicular pain is increasingly taught to undergraduate health professionals, and it is pain associated with nerve root compression or irritation. It has the characteristic quality of sharp, shooting pain in a relatively narrow band, known as a dermatome. Dermatomes are mapped out and can be used to determine the segmental level of the nerve root compromise. Dermatomes themselves are variable both between and within individuals—they move in line with shifting receptor fields of spinal neurons.^{3, 4} (Fig. 6.2). Nonetheless, current convention tells us that if the radicular pain is associated with compression of the nerve root, it will usually be accompanied by neurological abnormalities, e.g. paraesthesia corresponding to a reasonably typical dermatomal distribution or muscle weakness corresponding to a reasonably typical myotome.

Irritation of the dorsal root ganglion can evoke a similar distribution of pain, although the shooting pain may be delayed for several seconds and will sometimes

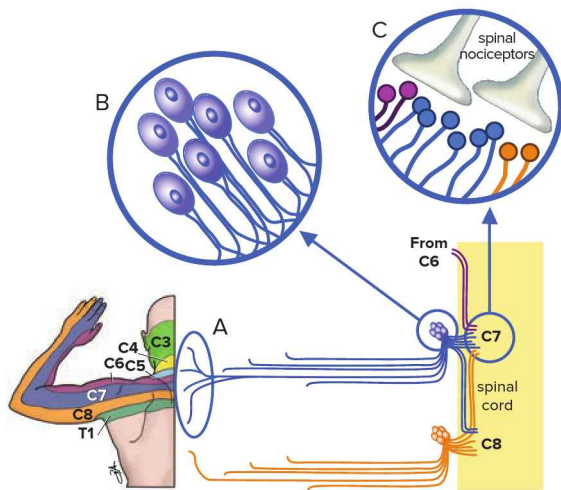


Figure 6.2 Radicular pain runs in well-characterised narrow bands according to which nerve root is affected. The left panel shows the dermatomal distribution of the neck region. The image also explains mechanisms that can underpin radicular pain—convergence of multiple incoming nociceptors. Note that radicular pain within the peripheral supply of a single spinal segment can occur via convergence of multiple branches of single nociceptors (A). Radicular pain can also occur via nociceptors within the dorsal root ganglion of that peripheral nerve (B). Pain can occur within the adjacent spinal nerve root territory via convergence within the dorsal horn, where projections from levels above (C6) or below (C8) can terminate alongside those from the spinal segment concerned (C)

reverberate (see Butler and Moseley 2013⁵ for more on this). That nerve root irritation and dorsal root ganglion irritation evoke pain in the area normally supplied by the affected nerve is intuitively sensible. The spinal neurons that convey the danger message have no method by which to differentiate where along the primary nociceptor the activity was generated, which leaves the brain allocating the danger to the target tissues.

Somatic pain

The other type of referred pain—the one that is largely ignored in traditional medical (i.e. MBBS, MD) teaching—is often called somatic referred pain. The mechanisms that underpin somatic referred pain are not as structurally simple as radicular referred pain. This is because the ‘disrupted transmission’ occurs within the central nervous system. This might happen at the dorsal horn in the spinal cord, in the thalamus or in the brain itself (Fig. 6.3). For example, visceral and musculoskeletal afferents converge in the case of left arm pain evoked by cardiac

problems. Similarly, immune mechanisms form functional connections with contralateral spinal nociceptors and as such might evoke ‘mirror pains’ at the identical site on the opposite limb.

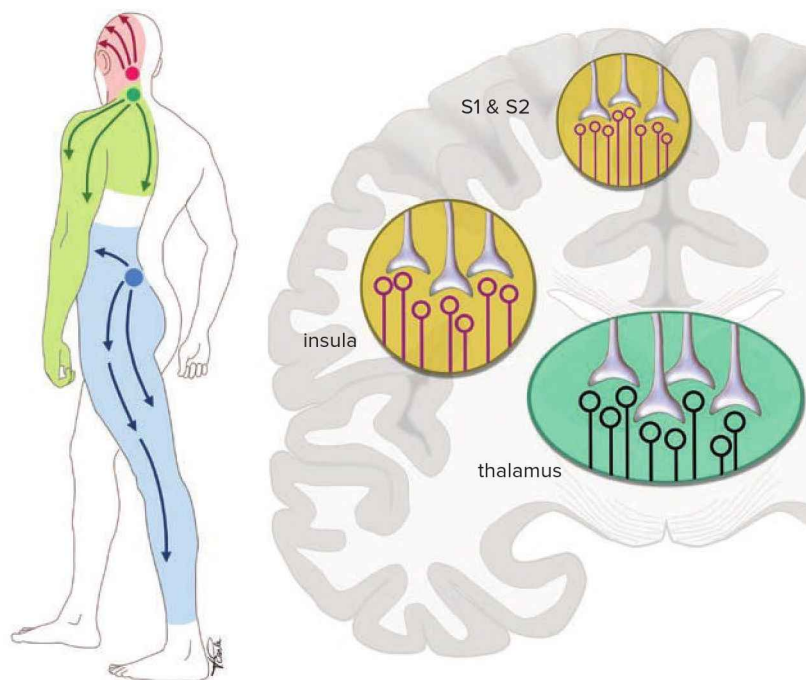
It can be helpful for the young clinician to conceptualise somatic referred pain as the brain attempting to localise the pain in response to ambiguous input (Fig. 6.3). As a result, the patient explains that the boundaries of pain seem indistinct, and that the pain can move. There is a large amount of variability between how individuals tell their story of this type of pain.

Two structures that are widely held by clinicians to be common contributors to somatic referred pain are (i) joints and (ii) myofascial trigger points (although the evidence concerning popular treatments for the latter is somewhat fragile). The growing ambiguity or *imprecision*, that seems to accompany somatic referred pain increases as the usual intracortical inhibitory mechanisms break down in persistent pain. The longer pain persists, the more ‘bodily real estate’ it tends to consume. Indeed, the attribution of danger to a particular area of *space*, in addition to *body*,^{6–8} is consistent with this loss of inhibition in brain areas dedicated to spatial localisation of body parts.¹⁰

Neuropathic pain

Neuropathic pain can be difficult to categorise and is defined as pain arising as a direct consequence of a lesion or disease affecting the somatosensory system.¹⁰ In contrast to pain that arises from activating nociceptive afferents (discussed above as pain with nociceptive contribution), pain can arise by activity generated without adequate stimulation of sensory nerve endings and is probably best described as a characteristic rather than a diagnosis. However, neuropathic pain can occur concurrently with nociceptive contributions to pain, that is, initial trauma can damage peripheral musculoskeletal tissue as well as neural tissue. This provides a mixed clinical picture that makes it more challenging for the clinician to unravel the contributor(s) to pain.

There is no gold standard or even clinical test for neuropathic pain. It represents clusters of features that may or may not have a history of injury or illness preceding the symptoms. Clinically, many find it useful to grade according to the level of certainty with which it can be determined in a patient¹⁰ (definite, probably, possible). This is based on the premise of pain distribution being consistent with dermatomes (that is, corresponding to peripheral innervation patterns or representations). Causes of neuropathic pain include diabetes, malignant disease, trauma and ischaemia. People presenting with pain (often to a musculoskeletal clinician) may have diabetes or an undiagnosed malignancy, so the clinician must remain alert to this possibility when a patient presenting with what would be expected to be



A

Figure 6.3 Somatic pain is an important type of input-dominated pain. As in the illustration of the person on the left side of the figure, pain can be referred in non-radicular patterns such as within a limb or 'quadrant'. Nociceptive signals converge in higher centres, e.g. thalamus, insula, and primary (S1) and secondary (S2) somatosensory areas.

input-dominated pain (e.g. local tissue trauma) reports ongoing pain with neuropathic features.

The clinical features of neuropathic pain may include burning or shooting pain, hypersensitivity, expansion of receptive fields, deep aching, motor deficits, abnormal neurological examination and spontaneous pain.^{2, 11, 12} A patient with neuropathic pain may experience both sensory loss (cold, warmth and pinprick) and the presence of certain hyperphenomena such as tingling or prickling sensations. However, clinical neurological testing may not be sensitive enough to detect the afferent (sensory) changes. There is no gold standard to identify neuropathic pain, nor consensus over whether neuropathic pain questionnaires should be used in clinical practice. Of the available questionnaires, PainDETECT and LANSS¹³ have the best clinical applicability as well as acceptable sensitivity and specificity.

CENTRALLY DOMINATED PAIN

These classifications merely aim to assist clinicians to form a treatment path. This is because people who have reduced thresholds to the way they perceive stimuli that is contributing to their pain (perhaps even a main driver) require individual approaches. The term 'centrally

dominated pain' refers to clusters of clinical features and includes sensory findings (both positive and negative findings). It is often thought of as pain that persists beyond usual time frames (however, this is difficult to quantify and could mean treatment was inappropriately directed) or as pain that is experienced without stimuli.

This is where it gets tricky as reduced sensory thresholds are also observed even in acute situations such as an ankle sprain. Thus, centrally dominated pain refers to the pain of a subgroup of people who have not responded to many traditional treatments. Unfortunately many people experience persistent pain and there are no tests that can identify people early that are at risk of persistent pain. The language strategies provided at the end of this chapter aim to help clinicians reduce their negative impact (and perhaps even have a positive impact) on their patient interactions. (Note that neuropathic pain has elements of both input-dominated and centrally dominated pain but it is included below.) Many conditions may have elements of neuropathic pain.

Central sensitisation

The expert sport and exercise clinician recognises that the relationship between pain and tissue damage becomes

more tenuous as pain persists.¹⁴ As pain persists, the central nervous system (CNS) switches to a state of spinal hypersensitivity, which generates allodynia (e.g. the experience of pain from non-noxious stimuli such as light touch) and the descending control pathways shift to excitation rather than inhibition.

Central sensitisation results from a complex cascade of physiological events involving all levels of the CNS and many areas of the brain¹⁵ (see Chapter 5 for more detail). Often the patient has a reduced pain threshold that is not explained by peripheral sensitisation. (Note that the clinical finding of lowered pain threshold can be explained by one of two mechanisms—central or peripheral sensitisation. This amplified neural signalling and potential for evoked pain via non-nociceptive channels elicits pain hypersensitivity and may represent a shift in the main contributor to pain. Neural impulses may be spontaneous.

These features can be difficult to clinically assess and have been mainly reported using equipment such as qualitative sensory testing, which is expensive and time consuming. Light touch testing (e.g. with von Frey filaments) and two-point discrimination testing require practice to obtain reasonable reliability and repeatability and also rely on the assumption that there are established normative data—which there are not! This may sound very negative, but it is important for clinicians to put everything together in the clinical picture. The clinician should be alert to the possibility of central sensitisation when pain persists beyond the ‘normal healing’ time frames for an injury. At the same time, please listen to the voice saying that you also need to watch for ‘red flags’ (masquerades—see Chapter 7) at that point too. Unfortunately, no simple clinical tests provide confirmation that the pain presentation is centrally dominated. The clinician needs to take a reading from a cluster of signs and symptoms.

Another clue that pain may be centrally dominated is that it may respond poorly to treatments such as manual therapy, because CNS changes may have altered neural function (e.g. a light touch may evoke pain).

After an injury (e.g. ankle sprain), a period of increased sensitivity is an adaptation that reduces the risk of secondary injury and provides an opportunity for tissue to heal;¹⁴ however, persistent hypervigilance can substantially impair quality of life. The concept of threat and pain is summarised in the previous chapter and also detailed in the book *Explain Pain*^{1,5} (Fig. 6.4). Remember that the relationship between perceived threat and pain is true of all pain presentations, not just chronic pain states.

Table 6.2 assists clinicians to interpret information obtained in their subjective assessment and to guide pain classification. It includes common assessment techniques

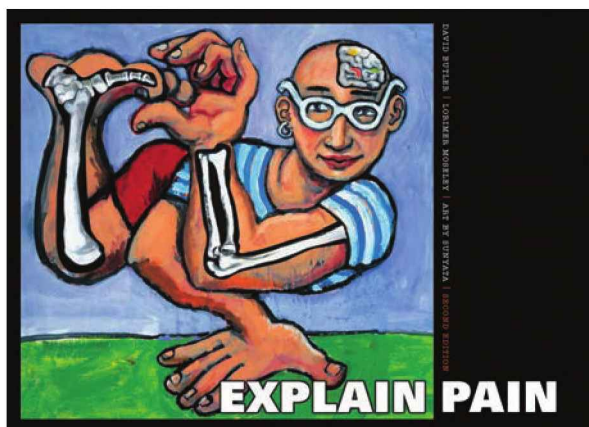


Figure 6.4 *Explain Pain* is a book by Dr David Butler and Professor Lorimer Moseley, but the term ‘explain pain’ has increasingly been used to refer to an approach to treating centrally dominated pain. We recommend it as part of the essential library for the *Clinical Sports Medicine* community
DAVID BUTLER AND G LORIMER MOSELEY, *EXPLAIN PAIN*, 2003, NOIGROUP PUBLICATIONS

and tools that may be used to assess sensory processing and pain.

It can be difficult to classify patients but some consideration of the primary contributors to their clinical presentation may assist with efficacious treatment selection. Figure 6.5 aims to assist clinicians with their clinical reasoning process. Of note, clinicians may be persistent in trying to identify a tissue-based diagnosis with ongoing pain. Conversely and of no benefit to patients, many are told the pain is all in their head! There are a number of patterns that can assist with using Figure 6.5. Here is a list of clinical symptoms and signs to assist with identifying centrally dominated pain presentations (refer to *Explain Pain* for greater detail):^{1,5}

- loss of two-point discrimination¹⁷
- language of pain—‘everything hurts’
- cold sensitivity
- tenderness to palpation away from injury site (secondary hyperalgesia)
- multiple diagnoses including myofascial syndrome, fibromyalgia
- red flags (see Chapter 7), including history of traumatic events, anxiety/depressive states
- cyclical or seasonal presentation (without links to changes in load or mechanical stimulus).

Table 6.3 includes common assessment techniques and tools that may be used to assess sensory processing and pain.

Table 6.2 Questions that clinicians can use to guide their approach to patients with pain. Note there generally exists a crossover among contributors to pain^{2, 11, 14, 16}

Question	Examples of responses	Clinical reasoning
Onset: was there an initial incident/injury and when was it?	Some people report a clear onset of pain, others do not Some individuals do not report pain even from a clear episode of severe trauma	Is the tissue a predominant driver of the pain? (No recollection of pain makes this less likely but not impossible) Overuse presentations (of input-dominated pain) may not have a sudden onset In the case of persistent pain, the incident may be a long time ago (may or may not be due to centrally dominated pain)
Where is the pain? Does it move around or spread?	Localised pain Within dermatomal distribution Pain that does not fit a dermatomal or myotomal distribution	May reflect nociceptive including compression of neural tissue May be referred (e.g. from the spine or viscera) Potential centrally dominated or neuropathic pain
Can you describe the symptoms?	Burning, electric shock-like, shooting, crushing, paraesthesia, and pins and needles Specific clusters of symptoms such as sweating, swelling, glossy skin, together with burning pain may reflect increased sympathetic activity	May reflect neuropathic pain (input dominated) These features are characteristic of complex regional pain syndrome. This is an example of centrally dominated pain
Does anything provoke the pain? (Examples given include visual and mechanical stimuli.)	No	If the patient has pain without a physical stimulus, consider centrally dominated pain such as neuropathic pain, rheumatological causes, sensitisation, inflammation or a sinister cause (see Chapter 7) Even being at rest may irritate some nociceptive pain (including neural irritation, for example, if your ulnar nerve has pressure when your elbow rests on something) People may report that pain swaps sides, or limbs or 'pain has a mind of its own' Pain may be unpredictable, non-mechanical and non-anatomical in response to movement Possible to have movement-related memory pain that is reinforced by cognitive emotional factors Some patients with centrally dominated pain experience pain with visual stimuli
Does the patient report hyperexcitability of response to stimuli?	Yes Pain with light touch (mechanical) Cold allodynia (thermal) Chemical stimuli Wind-up-like pain (abnormal temporal summation, that is, the perceived increase in pain intensity over time when the stimulus is repeated above a critical rate) or aftersensations (persistence of pain after termination of a painful stimulus)	Possibly centrally dominated or neuropathic pain. Chemical and cold allodynia can be difficult to assess in the clinic. Patients can report hypersensitivity to cold Described following repetitive noxious or non-noxious stimuli (like light touch can evoke pain)

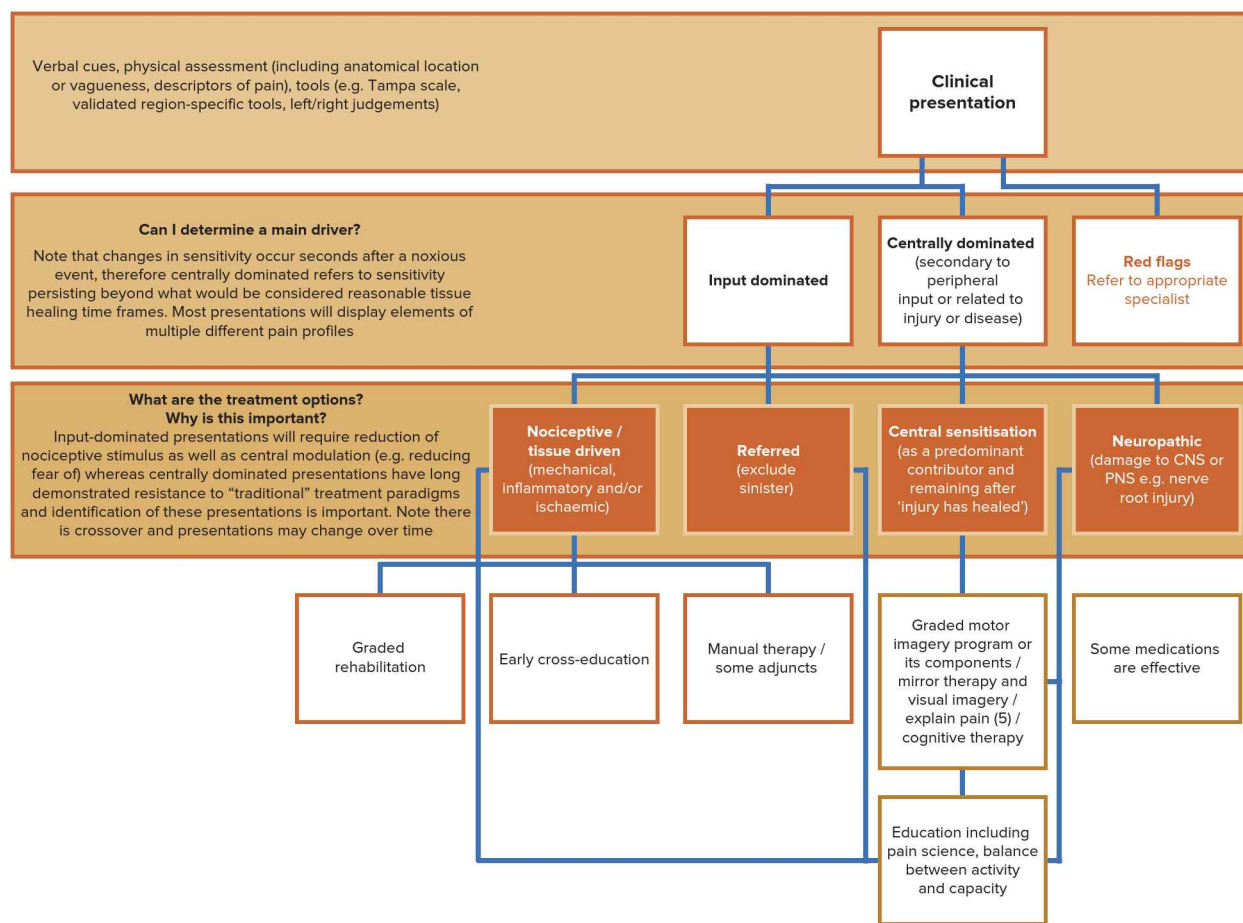


Figure 6.5 Flowchart to assist with clinical presentation and treatment path

MOTOR ADAPTATION TO PAIN

It is important for the clinician to appreciate that there are movement changes with pain—technically, ‘motor adaptations.’ Understanding what may underpin abnormal movement can allow appropriate rehabilitation (see Chapter 18).

Clinicians will recall examples of what pain scientists call the *pain-spasm-pain model*—pain increases muscle activity which in turn causes pain.²¹ Consider secondary muscle spasm associated with some forms of back pain. In contrast, the *pain-adaptation model* postulates that pain actually reduces activation of agonist muscles and increases activation of antagonist muscles²² to limit movement and avoid provoking pain and further damaging tissues. An example of this would be inhibition of the quadriceps to limit patellar movement in cases of patellofemoral pain.

Both of these types of motor change may reduce movement variability and that abnormal pattern may in turn increase the likelihood of recurrence of

pain.^{14, 20, 23, 24} Importantly, a so-called ‘protective movement-related pain memory’ may even hamper some attempts at rehabilitation for some patients.^{1, 25}

Limiting movement can be effective in reducing pain in the short term and as long as there is a perceived threat, motor outputs may serve to protect the part. These changes in motor learning occur at multiple levels of the motor neuroaxis (e.g. motor unit, spinal cord, brain), so the patient may need to relearn appropriate motor patterns (i.e. reprogram motor outputs) after a period of sustained protection. The expert clinician builds this in to the progressive rehabilitation and exercise prescription (see Chapter 18).

Treatment options for patients with pain

CNS modulation occurs with all sensory input—memories, thoughts, fears and expectations will influence and shape the output. There are many ways to treat people with centrally dominated pain and many aspects of the

Table 6.3 Tools for sensory processing and pain reporting

Clinical manifestation	Definition	Clinical examples	Tools
Allodynia	Stimuli that do not normally evoke pain (non-noxious stimuli) are now capable of evoking pain	Jumping can be painful for someone with patellar tendinopathy. Walking may be painful for someone who has sprained their lateral ankle	Observation Physical assessment, including palpation Note that allodynia is a normal and protective response to injury, but can also be part of persistent pain states
Hyperalgesia (primary and secondary)	Stimuli that are normally painful now evoke more pain Lowering of pain threshold response to noxious stimuli	Primary hyperalgesia—peripheral sensitisation Secondary hyperalgesia—attributed to central sensitisation	Pressure pain algometer to detect mechanical hyperalgesia Qualitative sensory testing can examine the presence of thermal and mechanical hyperalgesia by checking thresholds and detection
Pain during function	Patient reports pain during activity Using a functional test enables re-testing of the task		Visual analogue scale Numerical rating scale (e.g. 0–10) Multidimensional descriptors such as McGill Pain questionnaire ¹⁸ Specific questionnaires for diagnosis, e.g. VISA-P for patellar tendinopathy ¹⁹
Disrupted motor imagery performance	Accuracy and time to recognise left and right body images	This test has three steps: 1. Initial spontaneous judgment 2. Mentally match image to mimic posture shown (requires intact working body schema/cortical representation) 3. Confirm or deny initial judgment. This is impaired in some conditions such as Complex Regional Pain Syndrome and phantom limb pain ²⁰	Android and iPhone apps can test this—for example Recognise

Be alert to the onset of centrally dominated pain

As an injury progresses from acute to chronic, the nature of the pain may change in some patients with centrally dominated pain becoming more likely as pain persists.

Let's take the example of an ankle sprain. In the acute setting with pain, swelling and disability, the major contributions to the pain are nociceptive input dominated.

In this example, the clinician has the benefit of the patient reporting that he or she had rolled the ankle, the presence of swelling in the area and perhaps their limping gait, to guide clinical assessment. The combination of subjective information and physical signs points to a tissue-based diagnosis, which helps the clinician formulate a management plan based on the philosophy that this is input dominated/nociceptive. Appropriate language that avoids fear mongering, and education to guide exercise and analgesic behaviours will be beneficial (or at least do no harm as the body heals).

If the person continues to limp after several months, is afraid to use the ankle, describes pain without activity or pain

with clothes against his or her ankle and perhaps there are skin changes, the clinician may start to think that the presentation has changed from input dominated to centrally dominated. In the former, the ligaments would be expected to be 'healed' after this period of time. However, there is no cut off for a transition to a centrally dominant picture (e.g. not greater than 3 months, as many texts will indicate).

The important point to make here is that manual techniques aimed at the ankle joint could easily make pain worse. The treatment path may need to be very creative and use techniques such as graded motor imagery to begin moving the virtual ankle before the physical ankle is re-introduced to rehabilitation. This is never easy, and the clinician must always remember the possibility of a missed diagnosis (such as a fracture) or a sinister cause, yet not continue to hunt for a tissue-based diagnosis when some contributors to pain cannot be scanned (such as beliefs and feelings).

treatment are different to that used for nociceptive pain (Fig. 6.5).

Education

Most patients with pain will benefit from accurate education about perceived threat and pain. The clinician must avoid potentially harmful language such as 'bone on bone' changes in osteoarthritis or 'massive disc rupture/herniation' in back pain.

Identification of centrally dominated pain is important because evidence suggests the best way to help patients lies in skilled practitioners and appropriate resources. In the future, strategies need to be improved that target all aspects of CNS changes, including cortical reorganisation.^{14,20} It is possible to achieve a positive outcome; ion channels can change in minutes (see²⁶ for review) and movement-specific motor maps can shift over days and even minutes.^{27, 28} This suggests that patients may benefit from rapid neurological changes which equate to symptom relief (usually less pain) as the nervous system is restored to a less sensitive state.

Manual therapy

Touch can have anti-nociceptive effects²⁹ and can also benefit patients in conjunction with exercise in some conditions such as shoulder disorders.³⁰ In mechanical neck pain, with or without headache, the Cochrane review concluded that manual therapy should only be done in conjunction with exercise and that mobilisation or manipulation alone was less effective.³¹ Inaccurate

statements from clinicians such as 'putting your spine back in place' should be avoided. Acupuncture has been shown to have short-term effects on pain.³²

The remaining chapters of the book will direct the reader to appropriate and evidence-based techniques. Given the potential for anti-nociceptive effects, it stands to reason this may be an appropriate part of management for someone that has nociceptive contributions to pain, but less so for someone demonstrating sensitivity (e.g. when light touch is painful).

Medications

Medications may modify pain. Examples include medications that block inflammatory drivers, blunt neural sensitisation or promote neural plasticity. This topic is covered in Chapter 17.

Exercise/progressive loading

Following the initial period of unloading after an acute injury (or after prolonged changes to movement as a protective strategy), it is critical for patients to move and reload tissue in a gradual manner that promotes progressive tissue adaptation. Rest reduces capacity³³ in musculoskeletal tissues. (See Mechanotherapy in Chapter 17).

In nociceptive presentations, the clinician may focus more on local tissue reloading whereas, when pain is more centrally dominant, the exercise program may need to drop to below that which provokes them and take a backseat to techniques such as graded motor imagery (GMI) and 'explain pain'. This is described in

Table 6.4 Language that may be unhelpful and how to rephrase it (adapted from *Explain Pain*—see text for more detail)⁵

Unhelpful language that may amplify pain experience or interfere with progress	Potential rephrasing that may be clinically appropriate	Considerations/rationale
'It's bone on bone in there.'	'Your MRI shows age- and load-appropriate changes.'	Asymptomatic pathology is common ^{36, 37} and focus on pathology (that may or may not be relevant for the patient's presentation) can be negative
'Your scan shows degeneration.'	'Your scan shows adaptations to age and load.'	Conversely, targeted education that demonstrates adaptation can improve outcomes ³⁸
'You have no core strength.'	'Your trunk muscles could be stronger.'	The body has remarkable ability to heal/adapt and, in many cases, to function better than prior to a medical incident. Treatment should facilitate this
'Your pelvis is out.'	'I am a complete idiot.'	OK—you can see Lorimer Moseley's influence here. But we are making a serious point and humour is a powerful communication tool, whether in textbooks or with patients

the GMI handbook where the GMI program is part of rehabilitation. When symptoms worsen, the techniques used are more focused on motor/functional empathy and implicit motor imagery such as left/right judgments, and as symptoms improve, the techniques are directed more towards mirror therapy and actual exposure to load. The start point can be established between the clinician and the patient by listening to their current activity and how well they tolerate it. Ultimately, clinicians should be assisting patients and athletes to restore function.

Graded motor imagery/explain pain/cognitive therapy/sensory discrimination training

Some people will not benefit from manual therapy or exercise as touch or movement may trigger their pain. In this instance, pain science education is essential. The goal of this approach is to help patients understand the concepts of threat and related body responses. Fundamental concepts include extinction of pain/signal and desensitisation with the aim of reducing the patient's feelings of helplessness.¹⁴

Some patients may benefit from the use of mirror therapy, visual distortion (e.g. minimising lens), graded motor imagery (GMI), and strategies that stimulate sensory and motor

representations with context.¹⁴ These methods are very well described in the book *Explain Pain* (see Fig. 6.6).⁵

Language

Language is a powerful tool and has potential for both negative and positive impacts.^{5, 34, 35} Skilful communication can better engage a patient in his or her own care decisions and this can improve outcomes. Table 6.4 provides some examples of language that may be considered unhelpful and how they can be rephrased.

SUMMARY

In many cases in sport and exercise medicine, physical assessment is aimed at establishing a structural diagnosis to guide an effective treatment plan in the hope that someone will fully recover. The expert clinician appreciates pain as a protective output, maintains a level of vigilance for when it is necessary to take a broader approach, assesses the various contributors to the patient's pain and considers secondary factors that may act to modulate it at the peripheral and/or central levels (see Fig. 6.3). This approach, which incorporates advances in pain science, will mean fewer patients are given inappropriate treatment that is missing an important element of their condition.

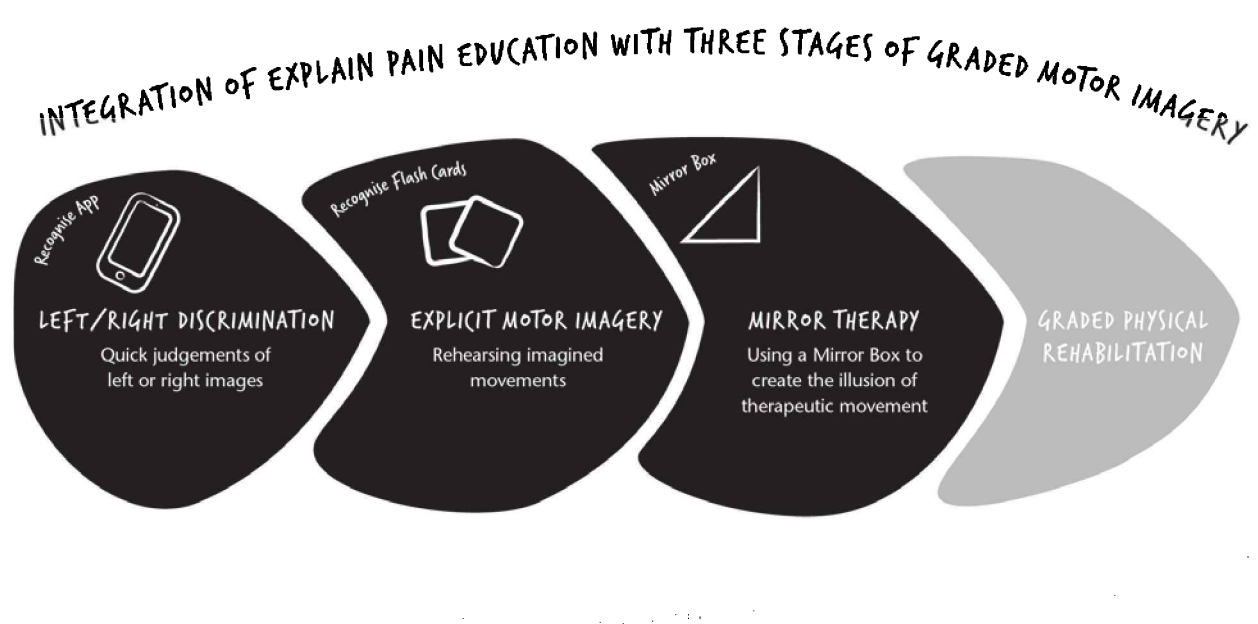


Figure 6.6 ‘Explain pain’ education, and the specific details of teaching graded motor imagery, is outside the scope of this book. We recommend that clinicians carefully consider this element when addressing painful conditions.

DAVID BUTLER AND G LORIMER MOSELEY, *EXPLAIN PAIN*, 2003, NOIGROUP PUBLICATIONS

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e